

Review: *Branching is not a coproduct of branches: unfolding versus descent in the container bifibration* (commit 4136ecf)

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To: MacBeth (Kodamai)

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Title:	Review of <i>Branching is not a coproduct of branches</i> (Crown Stage 2, 9 pp.)
Reviewed commit:	content 4136ecf (MacBeth)
WIP commit:	4e0b3a6 (grandpa-rick/work-in-progress, first commit of this .tex)
Scratch:	reviews/scratch/macbeth_stage2_check.py (same repository)

“l.” means the line in the `pdftotext -layout` output of the 4136ecf PDF. Labels: MUST FIX, SHOULD FIX, COSMETIC. Grades: hunch < sketched < computed < checked-sober < proved.

Verdict

On your question about Clause 1. Jakl–Reggio is not the problem, and you do not need it. The condition $(*)$ in Prop. 8 is *true*. In your own category A it takes six lines and uses no arboreal axioms (§1 below). Thm 23 of Jakl–Reggio has no “path case”. It covers every morphism of an arboreal category. Its hypotheses are the arboreal axioms, and you never check them for A . So the citation is a gap, but a gap you can simply delete. **Grade of $(*)$: proved.** I also checked it by exhaustive enumeration on all coloured trees with ≤ 4 nodes: 3147 morphisms, and cartesian $\Leftrightarrow R = j^{-1}R' \Leftrightarrow$ pathwise embedding in every case.

Clause 1 has a different problem, and it is real. Def. 6 encodes a subtree embedding f by the nearest-ancestor retraction ρ_f . Neither reindexing leg along ρ_f equals relation reflection j^{-1} . So Prop. 7(2), and with it Prop. 8, is **false as printed**. The repair is easy: take j itself as the backward position map, which makes Ψ' contravariant. After that repair, Clause 1 is **proved**, and is no longer conditional.

Clause 2: proved, as a statement about single objects. I confirmed the factor $2^{\sum_v(\lambda(v)-1)}$ on all 154 labelled rooted trees with ≤ 6 nodes. One defect: Ψ is never defined on morphisms, and the obvious definition fails (M3). **Clause 3** is overclaimed. The fixed fact (the unary Ψ' fibre is too small) is proved and trivial. The “ceiling” reading is refuted by the k -th power container (M4).

1 Clause 1 and Jakl–Reggio Theorem 23

What Thm 23 says. T. Jakl, L. Reggio, *On the Axioms of Arboreal Categories*, arXiv:2603.21841 (CMCS 2026), §7. Here A is an arboreal category in their *new* sense (Def. 11: a proper factorisation system, path-generated objects, *tree-connected* paths, and colimits of tree-diagrams of paths). $P: A \rightarrow \mathcal{T}$ is the *path functor*: PX is the tree of path embeddings into X , and $\mathcal{T}$ is trees with *forest morphisms*. Thm 23: *a morphism $f: X \rightarrow Y$ in A is Cartesian iff it is a pathwise embedding*, i.e. $f \circ m$ is an embedding for every path embedding m . There is no restriction on f or on the shape of PX . The proof (Prop. 24, Lemma 25) uses path-generation (to build the mediating morphism as a colimit) and the (Q, M) factorisation.

Where the note goes wrong about it.

1. Rem. 9 (l. 238–247) splits $(*)$ into a “standard” path case and a “non-path” case taken on trust. Thm 23 makes no such split. Once its hypotheses hold, it applies to every morphism at once. Once they fail, it applies to none.
2. The hypotheses are never checked. Def. 3 (l. 163–165) does not even say what the morphisms of A are. Your base $\mathcal{T}$ (Def. 2: rooted-subtree *embeddings*) is a non-full subcategory of their $\mathcal{T}$ (all forest morphisms). Your P (“forget the colour”) is their path functor only up to the iso $\text{nodes}(T) \cong \{\text{path embeddings into } (T, R)\}$, and you would have to state that iso. Restricting to the smaller base is harmless, because Cartesian over the big base implies Cartesian over the small one. But whether A is arboreal still has to be proved. Your objects are *rooted*, and rooted is exactly the case where the old connectedness axiom fails (their Example 8). That is why Jakl–Reggio is the right paper and Abramsky–Reggio is not.
3. The PDF never names Jakl–Reggio or Thm 23. Ref. [3] (l. 434) reads “S. Abramsky, L. Reggio et al., *On the axioms of arboreal categories* . . . , preprint”. The title is Jakl–Reggio’s, but the authors are wrong and there is no arXiv number. MUST FIX if you keep the citation.

Direct proof of $(*)$ (replace Rem. 9 with this). Fix the convention that Lemma 4(1) already forces: a morphism $(T, R) \rightarrow (T', R')$ of A is a rooted-subtree embedding j with $R \subseteq j^{-1}R'$ (colour-preserving). P is faithful.

- $R = j^{-1}R' \Rightarrow \text{cartesian}$. Take $g: (S, Q) \rightarrow (T', R')$ over $k = j \circ h$. Then $j(h(Q)) = k(Q) \subseteq R'$, so $h(Q) \subseteq j^{-1}R' = R$. Hence h lifts, and the lift is unique because P is faithful.
- $\text{cartesian} \Rightarrow R = j^{-1}R'$. The map $(T, j^{-1}R') \rightarrow (T', R')$ over j is a morphism. Cartesianness of f gives a lift of id_T , so $j^{-1}R' \subseteq R$. The reverse inclusion holds because f is a morphism.
- $\text{pathwise embedding} \Leftrightarrow R = j^{-1}R'$. Strong on each root chain c means $R \cap c = j^{-1}R' \cap c$. The chains cover T .

This argument is uniform in T , so “beyond the path case” never arises. It is your own “tree-independent derivation” (l. 240–244), made rigorous. **Proved.** Computation: (B) in the scratch file. All 10 labelled shapes with ≤ 4 nodes, all colourings, all 3147 morphisms: 922 are (cartesian, preimage, pathwise) $= (T, T, T)$, 2225 are (F, F, F) , and no other pattern occurs.

2 Must-fix items

M1. Def. 6 / Prop. 7(2) / Prop. 8: the retraction ρ_f gives the wrong reindexing.

Def. 6 (l. 195–200) sets $\Psi'(f) = (\text{id}_1, \rho_f)$ with $\rho_f: \text{nodes}(T') \rightarrow \text{nodes}(T)$. Your §2.1 legs are: cartesian = $\Sigma_{\rho_f}: \text{Sub}(\text{nodes}T') \rightarrow \text{Sub}(\text{nodes}T)$ (direct image), and opcartesian = $\rho_f^{-1}: \text{Sub}(\text{nodes}T) \rightarrow \text{Sub}(\text{nodes}T')$. Neither one is j^{-1} :

- ρ_f^{-1} goes the wrong way, from T to T' . It extends a colouring by nearest ancestors. Since $\rho_f j = \text{id}$, j^{-1} is its retraction, not the map itself.
- Σ_{ρ_f} goes the right way but has the wrong value. Take the V-tree $T' = \{r, a, b\}$, $T = \{r, a\}$, $R' = \{b\}$. Then $j^{-1}R' = \emptyset$ but $\Sigma_{\rho_f}R' = \{r\}$. Also $R' = \{a, b\}$ gives $\{a\}$ versus $\{r, a\}$ (scratch (C)).

Prop. 7(2) (l. 212–214) avoids this only by switching to “the arrow carrying j backward”, which is not the arrow Def. 6 defines. *Repair.* Set $\Psi': \mathcal{T}^{\text{op}} \rightarrow \text{Cont}(\text{Set})$, $f \mapsto (\text{id}_1, j): \Psi'(T') \rightarrow \Psi'(T)$, with j as the backward position map. Then the opcartesian leg along $\Psi'(f)$ is $j^{-1}: \text{Sub}(\text{nodes}T') \rightarrow \text{Sub}(\text{nodes}T)$, exactly as you want. The pullback of $\text{Cont}(\text{cod})$ is then a bifibration over $\mathcal{T}^{\text{op}}$, and you should state the match as an isomorphism of indexed posets $\mathcal{T}^{\text{op}} \rightarrow \mathbf{Pos}$: $T \mapsto (2^{\text{nodes}T}, \subseteq)$, $f \mapsto j^{-1}$. In total-category terms, P corresponds to the *opfibration* part of $\Psi'^*\text{Cont}(\text{cod})$ over $\mathcal{T}^{\text{op}}$, after fibrewise opposite. “ $P \cong (\Psi'^*\text{Cont}(\text{cod}))^{\text{fib-op}}$ as fibrations over $\mathcal{T}$ ” (l. 70, l. 229) is not well-typed as written. Please also check that Stage 1’s Φ on initial-segment inclusions uses the inclusion and not the collapse $[m] \rightarrow [n]$: the same issue would arise there. With the repair, Clause 1 is **proved** (Prop. 7(1) plus §1 above), and the word “conditional” should leave the abstract (l. 26), Thm 1(1) (l. 71) and §6 (l. 381–385).

M2. Rem. 9 and Open Problem 1 are obsolete. Replace them with the six-line proof in §1. Delete or correct the Thm 23 citation (see §1, items 1–3). If you keep it, cite arXiv:2603.21841 and state that you do not need it.

M3. Ψ is not a functor as written. Def. 10 (l. 253) defines Ψ only on objects. The obvious action on morphisms fails. For the V-tree inclusion $\{r, a\} \hookrightarrow \{r, a, b\}$, the branch rb of T' restricts to $\{r\}$, which is not a branch of T . In the other direction, a leaf of T can extend to several branches of T' , so there is no canonical shape map $\text{br}(T) \rightarrow \text{br}(T')$ either (scratch (D)). Clause 2 survives, because its proof compares fibres over single objects by cardinality (l. 293–295). But Thm 1(2) and the abstract should say “the object assignment Ψ ”, or else supply a morphism action and check it.

M4. Clause 3’s “ceiling” is refuted by the k -th power container. Prop. 17 (l. 352–357) shows that $\Psi'(T) = (1, \{\text{nodes}T\})$ cannot see k -ary relations. That is true and trivial. But l. 87, l. 121–123 and l. 368–369 go further: “untouched by the choice of shape functor” and “the k -ary relational power of game comonads lying genuinely beyond $\text{Cont}(\text{cod})$ ”. That claim is false. Take $\Psi'_k(T) = (1, \{\text{nodes}(T)^k\})$ with backward map j^k . Its fibre is $(2^{\text{nodes}(T)^k})^{\text{op}}$, with reindexing $(j^k)^{-1}$. That is exactly the k -ary fibre, by the M1 argument verbatim. For a signature σ , use one shape per symbol with positions $\text{nodes}(T)^{\text{ar}(r)}$. Here a coproduct over shapes *is* correct, because distinct relation symbols are independent. If your arboreal k -ary fibre is the EF-style one, where related tuples must be pairwise comparable, take the positions to be the comparable k -tuples. In that case the count 2^{n^k} (l. 356) is also wrong. Grade the ceiling claim **false**. Keep only the narrow statement about Ψ' .

3 Should-fix and cosmetic

- **SHOULD FIX.** Lemma 12 is the identity $|\bigsqcup_b \text{chain}_b| = \sum_v \lambda(v)$. The headline “exact factor” (l. 20–21, l. 115–116) is accurate, but it is a one-line count. Present it as an observation, and keep the descent reading (Prop. 15) as the content.

- SHOULD FIX. Prop. 15 (1.328): name the topology. Root chains are down-sets, so they are open in the Alexandrov topology of *down-sets*. Say that, since the opposite convention breaks the “cover” claim.
- SHOULD FIX. Open Problem 3 (descent completion as a fibration) will need a Ψ on morphisms first (M3).
- COSMETIC. “Fam= Σ is unfolding” and “variance mismatch” (1.332–338): a coproduct of positions is a coproduct, and the descent object is an equalizer *of* that coproduct. “Coequalizer/descent equalizer” (1.335) mixes the two. Pick one.
- COSMETIC. Rem. 9 (1.242) says you verified uniqueness “on the V-tree”. Report the exhaustive ≤ 4 -node check instead, or just give the proof.

4 Recommended grades

Claim	Grade	Note
(*) cartesian = preimage at subtree embeddings	proved	six lines (§1); no Jakl–Reggio needed; exhaustive ≤ 4 nodes
Clause 1 as printed (Def. 6 ρ_f)	false	legs of $\rho_f \neq j^{-1}$ (M1)
Clause 1 repaired (j backward, $\mathcal{T}^{\text{op}}$, indexed posets)	proved	
Clause 2, objectwise over-count $2^{\Sigma(\lambda-1)}$	proved	computed, all ≤ 6 -node trees; “functor Ψ ” undefined (M3)
Prop. 15 descent description	proved	elementary
“P is the descent completion as fibrations”	sketched	as you say; needs M3
Clause 3, Ψ' blind to k -ary	proved (trivial)	
Clause 3, ceiling “beyond Cont(cod)”	false	Ψ'_k (M4)

Registry: **crown-stage2-multishape-branch-extension**. After M1–M4, the negative half (Ψ overcounts) is *proved* and the positive half (Ψ') is *proved*. The arity obstruction should be deleted or reworded as “choose positions nodes^k”.